MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER

STATISTICS

MG5SECSTA300 – Applied Statistics and Data Mining

Duration: 1.5 Hour Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions
Answer any seven questions. Each question carries 2 marks.

1. What are the different types of Index numbers? [R][1]
2. Define Kelly's index number. [R][1]
3. Distinguish between simple AM and simple GM index numbers. [U][1]
4. Define Chain base index number.[R][1]
5. What are the sources of Vital statistics in India. [R][2]
6. Write limitations of Total fertility rate. [U][2]
7. Define Gross reproduction rate and Net reproduction rate. [R][2]
8. What is Market basket analysis? [R][3]
9. What is noise in data, and how does it differ from an outlier? [An][3]
10. Distinguish between Additive and Multiplicative models of Time series. [An][3]

Part B

Short Essay questions
Answer any four questions. Each question carries 6 marks.

11. Define Factor reversal test. Examine whether Paasche's and Fisher's index number satisfies this test. [An][1]
12. Define Cost of living index number. Mention its uses and limitations. [U][1]

13. Briefly explain any 3 measures of Fertility rates.[U][2]
14. Explain the underlying assumptions of a life table and give a brief description of its construction. [U][2]
15. Explain the four components of a Time series with an example of each.[U][3]
16. Discuss how data mining is used in Healthcare, Finance, Marketing and Social Sciences. [U][4]

Part C

Essay questions

Answer any one question. Each question carries 12 marks.

17. Construct Laspeyre's, Paasche's and hence Fisher's index number for the following data: [A][1]

Commodity	Base year Price	Base year Expenditure	Current year Price	Current year Expenditure
A	6	48	10	60
B	4	20	6	120
C	10	60	12	96
D	8	200	14	336

18. For the distance matrix D , Apply complete linkage method and draw the corresponding dendrogram.[A][3]

$$D = \begin{bmatrix} 0 & 9 & 3 & 6 & 11 \\ 9 & 0 & 7 & 5 & 10 \\ 3 & 7 & 0 & 9 & 2 \\ 6 & 5 & 9 & 0 & 8 \\ 11 & 10 & 2 & 8 & 0 \end{bmatrix}$$